

Consumer Preferences in the Used Vehicle Market

Willingness to Pay for Powertrain Type, Driving Range, Vehicle Condition, and Operating Cost

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Why the used-vehicle market?

- Used vehicles are **~70% of all U.S. vehicle transactions** — but EV research has focused almost entirely on **new-vehicle buyers**
- The used-BEV market is growing fast, and it matters beyond its own transaction volume:
 - Extends vehicle and battery lifetimes, supports a circular economy
 - Lowers the effective cost of EV ownership for lower- and middle-income households
 - Underpins new-vehicle residual values (which affect new-EV leasing economics)
- Used BEVs face **extra** friction new EVs don't: uncertainty about prior condition/reliability, and battery aging that odometer + age alone don't fully signal

Research questions

This study estimates willingness to pay (WTP) for vehicle attributes among prospective used car/SUV buyers, and asks:

1. **How do WTP estimates for vehicle attributes (powertrain, range, age, mileage, operating cost) compare across vehicle segment and budget tier?**
2. **How do existing used BEVs compete against their conventional or hybrid counterparts as vehicles age, in the real market?**

Survey & choice experiment

- 2,254 respondents planning a used car/SUV purchase within two years (Dynata + Prolific)
- Each completes 6 choice tasks: pick among 3 hypothetical used vehicles, or "none of these"
- Attributes varied: **powertrain** (ICEV/HEV/BEV), **price**, **operating cost**, **driving range** (BEV only), **age**, **mileage**
- Design stratified by **vehicle body type** × **budget tier** (4 versions), so attribute ranges match a plausible purchase context

Question 1 of 6

Assume you are able to purchase a vehicle that looks like the one you selected (see photo). Which of the following versions of that vehicle would you be most likely to purchase? Please compare the key features shown for each option before making your selection.



If these were your only options, which would you choose?

Option 1	Option 2	Option 3	Option 4
Purchase price: \$ 40,000	Purchase price: \$ 45,000	Purchase price: \$ 40,000	Even if these were my best options, I would not choose any of these vehicles
Powertrain:  Conventional	Powertrain:  Gas hybrid	Powertrain:  Battery electric 200 miles on a full charge	
Model year: 2019	Model year: 2023	Model year: 2023	
Mileage: 20,000	Mileage: 25,000	Mileage: 25,000	
Operating cost: 15 cents per mile (25.7 MPG equivalent)	Operating cost: 9 cents per mile (37.9 MPG equivalent)	Operating cost: 9 cents per mile (97.2 MPG equivalent)	

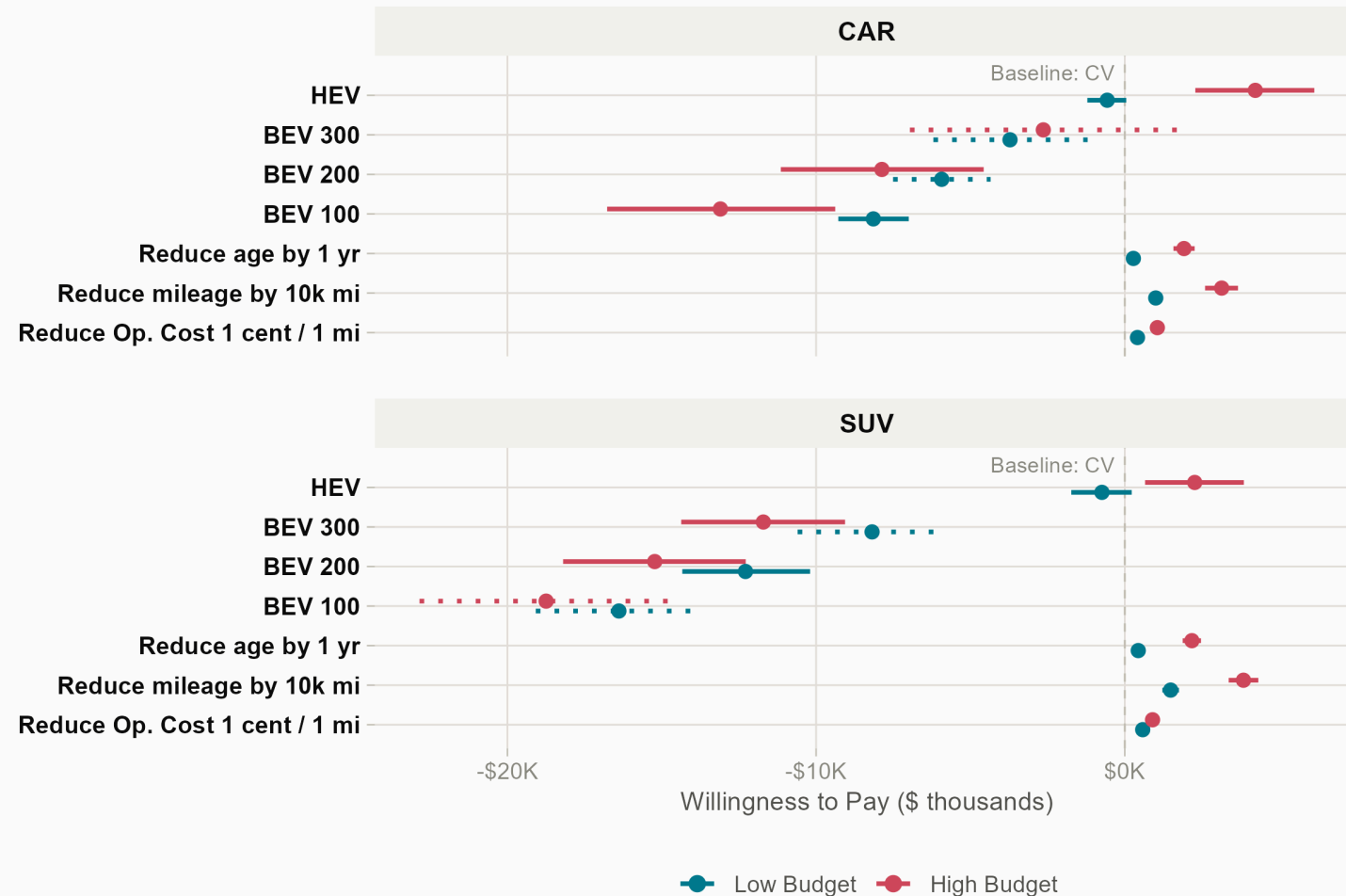
Sample & modeling strategy

- Six **mixed logit (MXL)** models estimated in **willingness-to-pay space**, stratified by:
 - **Vehicle segment:** Car vs. SUV
 - **Budget tier:** Low ($\leq \$20K$) vs. High ($> \$20K$)
- Panel-corrected (6 repeated choices per respondent), 5,000 Sobol draws, 10 multi-starts per model
- Why mixed logit, not plain multinomial logit?
 - Avoids the implausible IIA assumption (BEV/HEV/ICEV aren't interchangeable substitutes)
 - Lets us test whether preferences are **genuinely heterogeneous** across respondents — and they are, for almost every attribute

Q1 — Willingness to pay for vehicle attributes

Willingness to Pay for Vehicle Attributes

By vehicle segment and budget tier; error bars show 95% confidence intervals



Powertrain findings

- **HEV:** high-budget buyers place a real premium on hybrids — ~\$4,200 (car), ~\$2,300 (SUV) — low-budget buyers show no premium
- **BEV:** penalty relative to conventional vehicles everywhere, but it **narrows sharply with range**:

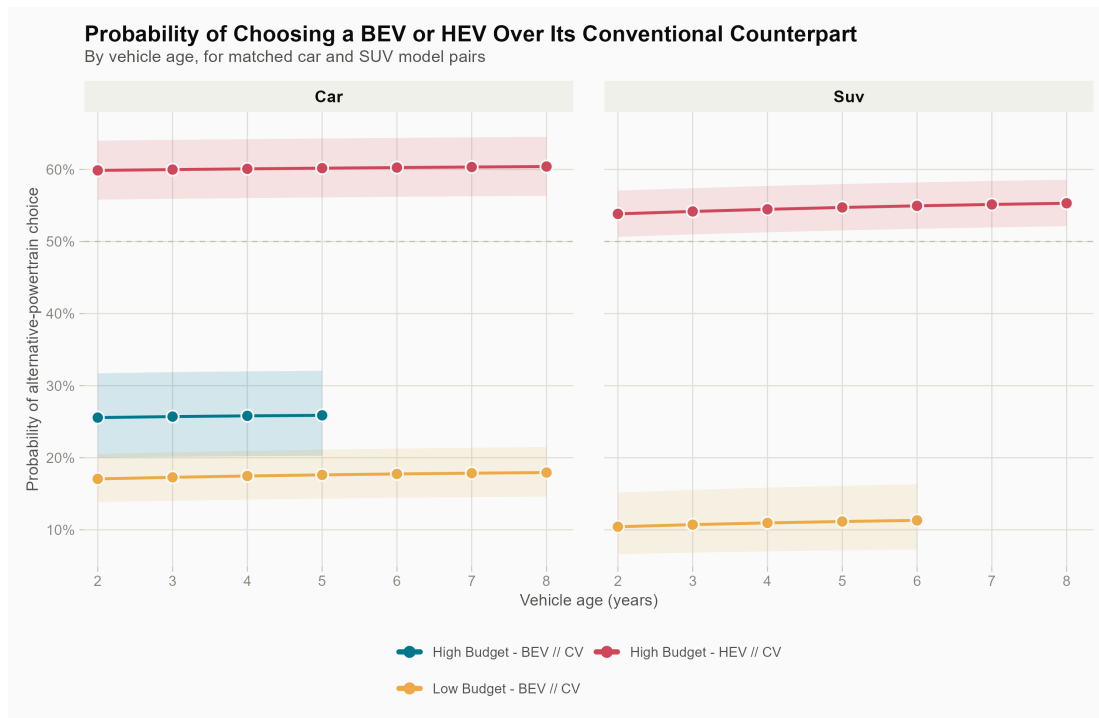
	100 mi range	300 mi range
Car, high budget	−\$13,100	−\$2,600 (≈ parity)
Car, low budget	−\$8,200	−\$3,700
SUV, high budget	−\$18,800	−\$11,700
SUV, low budget	−\$16,400	−\$8,200

Condition attributes: age, mileage, operating cost

- All three penalize used-BEV value, and **penalties scale with budget tier** (high-budget buyers show larger dollar penalties for the same attribute)
- **Mileage is the largest penalty** of the three: −\$1,000 to −\$3,800 across segments/tiers
- **Vehicle age**: −\$300 to −\$2,200
- **Operating cost**: −\$400 to −\$1,100 (smaller than age for high-budget buyers)
- **SUV buyers show systematically larger penalties than car buyers** at the same budget tier for BEV, mileage, and age — operating cost reverses at the high-budget tier (car penalty larger there)

Q2 — Do real used BEVs compete with conventional vehicles?

- **Market-share simulation** (Helveston et al. 2014 approach): predicted choice probability between a real used BEV/HEV and its closest matched conventional counterpart, ages 2–8 years
- 5 matched real-market vehicle pairs (3 car, 2 SUV)



Market-share takeaways

- **Two HEV-vs-conventional pairings already exceed 50% share at every observed age:** high-budget car (~60%), high-budget SUV (~54–55%)
- **No BEV-vs-conventional pairing approaches parity at any age:** 26% (car, high budget), 17–18% (car, low budget), 10–11% (SUV, low budget)
- Predicted share is **nearly flat with vehicle age** across all five pairings (≤ 2 points of movement) — faster real-market BEV/HEV price depreciation roughly offsets the age penalty from the choice model, rather than closing the gap over time

Bottom line: hybrids are already market-competitive used-vehicle substitutes; BEVs are not yet, regardless of age.

Policy implications

- The federal used clean-vehicle credit (\$25E, up to **\$4,000**/30% of price) only closes the gap for buyers **already closest to parity** (high-budget car buyers, long range) — it's a fraction of the \$8K–\$19K gap facing most other segments
- The **\$25,000 price cap** on that credit excludes many of the 250–300-mile-range used BEVs that are closest to competitive, while subsidizing shorter-range vehicles the credit can't make competitive anyway
- **~\$15,000–\$20,000** of the age-related WTP gap exceeds normal depreciation — plausibly a **battery state-of-health information gap**, addressable via disclosure/certification/warranties rather than cash transfers
- **Segment-aware policy** — and continued promotion of HEVs, which are already market-competitive — looks more efficient than uniform subsidies

Conclusion

- **Driving range and vehicle age are the dominant value determinants** for used BEVs — effects that swamp budget tier or vehicle segment alone
- Used BEVs reach price parity with conventional vehicles in **only one segment**: high-budget car buyers, 300-mile range
- **Hybrids are already there** — matching or beating conventional vehicles in predicted market share for high-budget car/SUV buyers, at every observed age
- For everyone else, the used-BEV discount required is large, predictable, and currently under-served by existing incentive design

Limitations

- Stated, not revealed, preferences — hypothetical bias possible (direction uncertain)
- Online panel sample may over-represent information-engaged consumers
- MXL assumes continuous normal heterogeneity — a latent-class model would test for discrete preference clusters directly
- Battery state-of-health was not varied in this DCE (covered in a companion battery-attribute DCE)

Questions?

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